Fuad Ahmed

fahmed22@mun.ca | St. John's, NL | LinkedIn | **Portfolio**

Education

Memorial University of Newfoundland **St. John's, NL, Canada**
Bachelor of Engineering Co-op – Mechatronics Engineering Class of 2028
Director of Co-op – Engineering Society 'A'

Experience

Solace Power **May 2026 –August 2026**
Advanced Research Electrical Engineering Intern

- Designed and integrated a 400W wireless drone charging system using a Class E topology
- Assembled and tested RF and Wireless Power Transfer Systems for complex aviation applications
- Modified high-voltage RF Systems for increased efficiency, thermal stability, and signal integrity
- Tuned, characterized and tested resonant power converter and RF Power transmission PCBs
- Board bring up and SMT soldering down to 0402 components

Michelin **September 2025 – December 2025**
Electrical/Automation Engineering Intern

- Programmed human-machine interfaces using Allen-Bradley PLC
- Prepared drawing packages and schematics for the installation of manufacturing machinery
- Conducted I/O testing and safety checks for sensors, cabinets and motors on new installations
- Designed and built a testing panel to power and control 480V and 120V motors

Memorial University Student Design Hub **January 2025 – April 2025**
Innovation & Prototyping Co-op

- Completed advanced training on CNC Milling
- Assembled, soldered, and tested microcontroller and power distribution PCBs
- Instructed students on 3D printing using Bambu Lab printers

Engineering Teams

Paradigm Engineering **September 2023 - Present**
Team Lead & Electrical Designer

- Leading a multidisciplinary team of 50+ students to develop a full-sized autonomous go-kart
- Designing a custom power distribution PCB to distribute power to all onboard components.
- Designing a custom STM32-based PCB to control all onboard motors, actuators and remote e-stop. This includes USB, UART, SWD and CAN peripherals
- Designing a 10kW drivetrain for an electric vehicle, including batteries, relays, and e-stops
- Raising over $45k in project funding through grants and sponsorships

Formula MUN **June 2026- Present**
High Voltage Systems Lead

- Leading the design and implementation of the tractive system for an FSAE EV car
- Designing and assembling a 250V battery pack with an 11-point shutoff circuit
- Designing and integrating an inverter and custom drivetrain for a 100km/h top speed

Skills

Software: KiCad, Altium Designer, Schematic Capture & Layout, LTSpice, Matlab, C++, Python
Electronics: Analog & Digital Circuit Design, PCB Design, Board Bring-Up, Soldering
Lab Testing: Oscilloscopes, Function Generators, Impedance Analyzers, Vector Network Analyzers